Bayesian Statistics I: Foundations

BSAD 8310: Business Forecasting — Lecture 10

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Lecture Outline

Why Bayesian Thinking?

Frequentist methods give point estimates with confidence intervals. Bayesian methods give full probability distributions over unknowns, encoding uncertainty explicitly.

Frequentist vs. Bayesian

Frequentist (Lectures 1–9):

- Parameters θ are **fixed but unknown**
- Probability $=$ long-run frequency of events
- A 95% confidence interval: "If we repeated this sampling procedure 100 times, 95 intervals would contain the true θ"
- Cannot say: "There is a 95% chance θ is in this interval"

Bayesian:

- Parameters θ are **random variables with distributions**
- Probability $=$ degree of belief, updated with evidence
- A 95% credible interval: "There is a 95% probability that θ falls in this interval, given the data we observed"
- Prior beliefs $+$ data $\rightarrow$ updated beliefs (posterior)

Business advantage: Bayesian forecasts come with full uncertainty distributions — not just a point estimate and a standard error. You can ask: "What is the probability that demand exceeds 10,000 units?" and get a direct answer.

Bayes' Theorem

Prior beliefs + likelihood of data = posterior beliefs.

Bayes' Theorem

Bayes' Theorem

$$\underbrace{p(\theta \mid \mathcal{D})}_{\text{posterior}} = \frac{\overbrace{p(\mathcal{D} \mid \theta)}^{\text{likelihood}} \cdot \overbrace{p(\theta)}^{\text{prior}}}{\underbrace{p(\mathcal{D})}_{\text{normalizing const.}}} \propto p(\mathcal{D} \mid \theta) \cdot p(\theta)$$

Prior $p(\theta)$ Beliefs about θ before data.

Likelihood $p(\mathcal{D} \mid \theta)$ Probability of data given θ.

Posterior $p(\theta \mid \mathcal{D})$ Updated belief after observing $\mathcal{D}$.

Business Example

Estimating conversion rate θ:

- **Prior:** $\theta \sim \text{Beta}(5, 45)$ (≈10% from past promotions)
- **Data:** 30 conversions / 200 exposures
- **Posterior:** updated θ distribution

Prior Distributions

The prior encodes what you know (or assume) before seeing data. Choose it to reflect genuine prior knowledge.

Common Prior Distributions

Beta distribution:

$$\theta \sim \text{Beta}(\alpha, \beta)$$

- Support: $[0, 1]$
- Prior for proportions (conversion rate, market share)
- $\text{Beta}(1, 1) = \text{Uniform}[0, 1]$
- α, β: counts of successes/failures

Normal distribution:

$$\theta \sim \mathcal{N}(\mu_0, \sigma_0^2)$$

- Support: $(-\infty, +\infty)$
- Prior for regression coefficients
- Weakly informative: $\mathcal{N}(0, 1)$ after standardizing
- Normal prior + Normal likelihood $\rightarrow$ Normal posterior

Exponential distribution:

$$\theta \sim \text{Exp}(\lambda)$$

- Support: $[0, +\infty)$
- Prior for positive quantities (variance, rate)
- $\text{HalfNormal}(\sigma)$ common for std. deviations

A **weakly informative prior** concentrates mass on plausible values without dominating the likelihood — better than flat priors (sampling problems) or overly tight priors (ignores data). (Gelman et al. 2013)

Posterior Inference with PyMC

When we cannot solve Bayes' theorem analytically, Markov Chain Monte Carlo (MCMC) samples from the posterior.

Markov Chain Monte Carlo (MCMC): The Big Idea

Analogy: Exploring a hilly landscape in the dark — you spend more time in high-probability areas. After many steps, visited locations trace the posterior.

The problem: $p(\mathcal{D})$ in Bayes' theorem is usually intractable for real models.

MCMC solution: sample from the posterior instead:

1. Start at initial $\theta^{(0)}$
2. Propose a new θ^* nearby
3. Accept θ^* with probability $\min\left(1, \frac{p(\theta^* | \mathcal{D})}{p(\theta^{(t)} | \mathcal{D})}\right)$
4. After many steps, the chain converges to $p(\theta \mid \mathcal{D})$

Metropolis Sampling

The simplest MCMC algorithm. PyMC uses NUTS (No-U-Turn Sampler) by default — adapts step size and exploits gradient information.

The posterior **samples** are the answer: compute means, medians, credible intervals, and event probabilities directly from the sample histogram.

PyMC: A First Model

Estimating a Conversion Rate

```
import pymc as pm
# 30 successes out of 200 trials
n_trials, n_success = 200, 30
with pm.Model() as model:
theta = pm.Beta('theta',
alpha=5, beta=45)
obs = pm.Binomial('obs',
n=n_trials, p=theta,
observed=n_success)
trace = pm.sample(2000,
return_inferencedata=True)
pm.plot_posterior(trace)
```

Reading the output:

- **Posterior mean:** best single estimate of θ
- **94% HDI:** shortest interval with 94% posterior mass (Bayesian credible interval)
- **Posterior histogram:** full uncertainty picture

PyMC models are *generative*: define priors, describe data generation, and the sampler does the rest.

Prior Predictive Checks

Before fitting the model to data, simulate predictions using *only the prior*. This checks whether the prior is sensible.

Prior Predictive Sampling

```
with model:
prior_pred =
pm.sample_prior_predictive(500)

pm.plot_prior_predictive(prior_pred)
```

If prior predictive samples are completely implausible (e.g., sales of 1 billion units), your prior is too wide — tighten it. If all samples look the same, it is too tight — widen it.

Prior predictive checks are **model debugging**. Run them before every Bayesian model to catch specification errors early. (McElreath 2020)

What to look for:

- Does the prior generate plausible data?
- No impossible values (negative sales, probabilities > 1)
- Not too concentrated (would override the data)

Key Takeaways

Bayesian inference: beliefs in, beliefs out. Data updates your prior into a posterior.

Lecture 10: Key Takeaways

1. **Bayesian inference** treats parameters as random variables with probability distributions. The posterior distribution $p(\theta \mid \mathcal{D})$ represents updated beliefs after seeing data.
2. **Bayes' theorem:** posterior $\propto$ likelihood $\times$ prior. The normalizing constant $p(\mathcal{D})$ is usually intractable, so we sample instead of computing it.
3. **Priors** encode prior knowledge. Use weakly informative priors (Beta, Normal, HalfNormal) to constrain parameters to plausible values without overwhelming the data.
4. **MCMC** (specifically NUTS in PyMC) draws samples from the posterior. From samples, compute any summary statistic: mean, median, HDI, or the probability that a parameter exceeds a threshold.
5. **Prior predictive checks** verify that your model generates plausible data before fitting. Run them every time.

Next: Bayesian Statistics II — Bayesian time series models, trend and seasonality, hierarchical models.

References I

- Gelman, Andrew et al. (2013). *Bayesian Data Analysis*. 3rd. Boca Raton, FL: Chapman and Hall/CRC.
- McElreath, Richard (2020). *Statistical Rethinking: A Bayesian Course with Examples in R and Stan*. 2nd. Boca Raton, FL: Chapman and Hall/CRC.